



RECURRENT HEMORRHAGIC GIANT CYSTIC INVASIVE BREAST CARCINOMA SUCCESSFULLY MANAGED WITH PREOPERATIVE TRANSARTERIAL EMBOLIZATION: A RARE CASE REPORT



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INTRODUCTION

Invasive breast carcinoma of no special type (IBC-NST) typically presents as a solid breast mass. Presentation as a **giant haemorrhagic cystic lesion is extremely uncommon** and may mimic a benign complex breast cyst. Although **transarterial embolization** has been described for haemorrhage from ulcerating or locally advanced breast cancers, its application for recurrent intralesional bleeding within a **cystic invasive carcinoma** has **rarely been reported**.



CASE REPORT

A 71-year-old postmenopausal woman presented with a **progressively enlarging painless right breast mass** of one year's duration. Ultrasonography demonstrated a large mixed solid-cystic breast lesion with a vascular mural solid component. Ultrasound-guided aspiration yielded 1,650 mL of **dark hemorrhagic fluid**, followed by core biopsy of the solid component, which confirmed grade 3 invasive breast carcinoma of no special type. Despite repeated aspiration, **rapid reaccumulation of hemorrhagic fluid** suggested ongoing bleeding from the vascular tumour component. **Transarterial embolization of multiple tumour-feeding arteries** was performed, achieving near stasis. The patient subsequently received neoadjuvant chemotherapy followed by right mastectomy with axillary dissection. Adjuvant radiotherapy was completed, with planned capecitabine therapy.



DISCUSSION

- Intracystic breast carcinoma is a rare entity, occurring predominantly in postmenopausal women.
- Complex cystic breast lesions have a significant risk of malignancy, particularly lesions with mixed solid-cystic components. Ultrasound in our case demonstrated a large complex cystic lesion with a vascular solid component.
- Although intracystic papillary carcinoma is commonly associated with cystic breast lesions, IBC-NST can rarely present as a giant cystic or hemorrhagic mass.
- Bloody or dark hemoserous aspirate, particularly when associated with a vascular mural/solid component, should raise suspicion for malignancy. Recurrent rapid reaccumulation after aspiration suggested ongoing tumour-related bleeding.
- Ultrasound-guided core biopsy of the solid component is essential for diagnosis. Aspiration alone may decompress the lesion but does not establish the underlying pathology.
- Transarterial embolization is rarely reported for intralesional bleeding in cystic breast carcinoma. In this case, embolization of multiple tumour-feeding arteries with PVA particles successfully controlled the recurrent haemorrhage and allowed subsequent neoadjuvant chemotherapy and definitive mastectomy.
- This case highlights the importance of early recognition of malignant features in complex cystic breast lesions and the potential role of embolization as a bridge to definitive oncological treatment.



Ultrasound shows large cystic lesion



Pre-embolization



Post embolization



CONCLUSION

This case illustrates a rare presentation of invasive breast carcinoma as a giant haemorrhagic cystic breast lesion with recurrent intralesional bleeding. Persistent haemorrhagic reaccumulation after aspiration should raise concern for malignancy. Preoperative transarterial embolization may represent a valuable adjunctive treatment for controlling tumour-related bleeding, allowing safe administration of neoadjuvant therapy and subsequent definitive surgical management.

References

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